Reaction Rates and Rate Laws

Rates, rate laws, units of k and first-order half-life

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$$\text{rate} = -\frac{\Delta[\text{reactant}]}{\Delta t} \cdot \text{rate} = k[\text{A}]^m[\text{B}]^n \quad \text{First order: } [\text{A}]_t = [\text{A}]_0\, e^{-kt} \cdot t_{1/2} = \frac{0.693}{k}$$

Give the units of every rate and every rate constant.

A. Understanding

1. Use collision theory to explain why (a) powdered marble reacts faster with acid than a single lump of the same mass, and (b) the rate increases with concentration.

B. Rates and stoichiometry

2. In $2\,\text{N}_2\text{O}_5\,(\text{g}) \longrightarrow 4\,\text{NO}_2\,(\text{g}) + \text{O}_2\,(\text{g})$, N_2O_5 is used up at 0.0060 M/s. Find the rates at which NO_2 and O_2 form.

3. The concentration of a reactant falls from 0.800 M to 0.560 M in 120 s. Calculate the average rate of its disappearance.

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C. Rate laws

4. The rate law for $2\,NO + 2\,H_2 \longrightarrow N_2 + 2\,H_2O$ is rate = k[NO]^2[H$_2$]. State (a) the order with respect to each reactant (b) the overall order (c) the units of k (d) the effect on the rate of doubling [NO].

5. For $2\,NO + O_2 \longrightarrow 2\,NO_2$: Experiment 1: [NO] = 0.020 M, [O$_2$] = 0.010 M, rate = 2.8×10^{-6} M/s. Experiment 2: [NO] = 0.040 M, [O$_2$] = 0.010 M, rate = 1.12×10^{-5} M/s. Experiment 3: [NO] = 0.020 M, [O$_2$] = 0.020 M, rate = 5.6×10^{-6} M/s. Determine the rate law and the value and units of k.

6. Using your rate law from question 5, what happens to the rate if [NO] is doubled and [O$_2$] is halved at the same time?

7. Give the units of k for a zero-order, a first-order and a second-order reaction (concentrations in M, time in s).

D. Half-life

8. A first-order reaction has $k = 2.31 \times 10^{-3}\ s^{-1}$. (a) Find its half-life. (b) If $[A]_0 = 0.400$ M, what is [A] after 900. s?

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E. Challenge

9. A first-order reaction has $k = 0.0150\ \text{min}^{-1}$ and $[\text{A}]_0 = 0.200\ \text{M}$. Calculate (a) [A] after 60.0 min and (b) the time for [A] to fall to 0.0500 M.

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